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Damage to Broca's Area:

A preliminary investigation into measuring degrees of aphasia following traumatic brain injury

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Abstract

Our research focuses on a group of 20 patients at Seaborg General Hospital who developed aphasia following traumatic brain injuries that affected Broca's area. We took fMRI scans of each patient, then did various cognitive tests while the patients were hooked up to EEGs. We found that

Methods

The subjects of this experiment range in ages from 21 to 78, with the median age being 45.5. 16 subjects (80%) were male, and 4 (20%) were female.

We took fMRI scans to confirm that each subject suffered a similar injury. We would disqualify patients who had more extensive damage that could affect cognition as it would interfere with the later results.

Following (https://www.researchgate.net/publication/347697311_Diagnosing_and_managing_post-stroke_aphasia) we ran a battery of tests on each participant while taking EEG data. We classified the data based on the severity of the injury (mild, severe, destructive) how much the aphasia affected the patient's day-to-day life (extremely disabling, moderately disabling)

Data

Sample 1: Extremely disabled 54-year-old female with severe injury

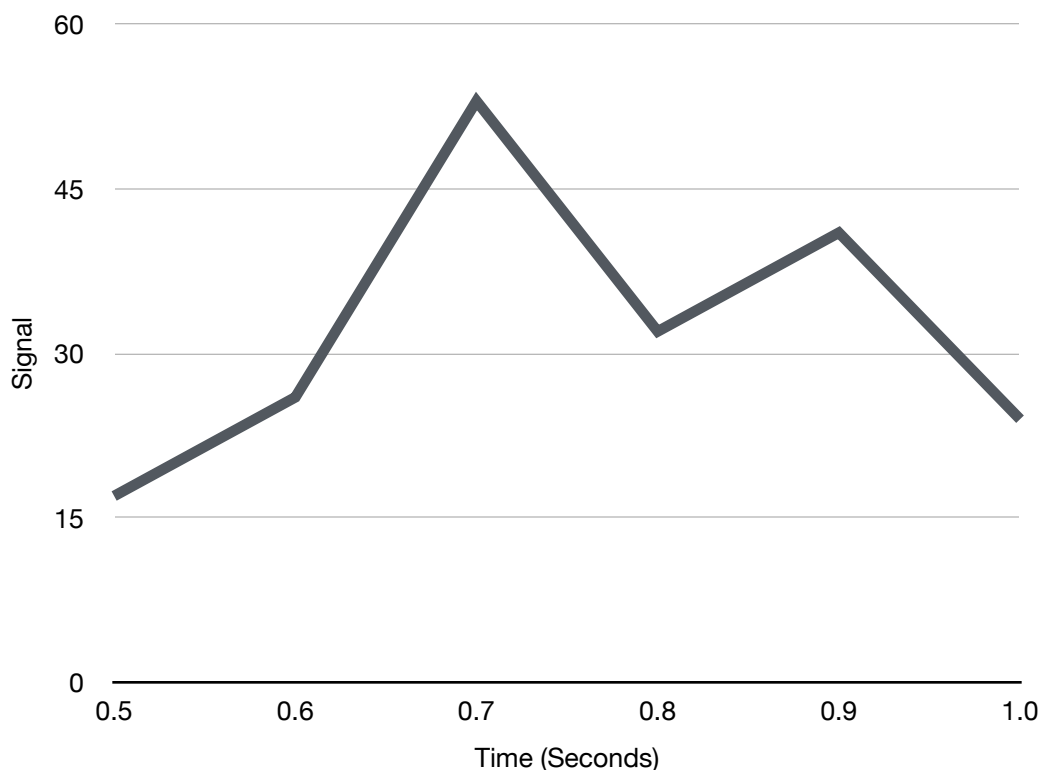


Figure 1

Sample 2: Moderately disabled 37-year-old male with severe injury

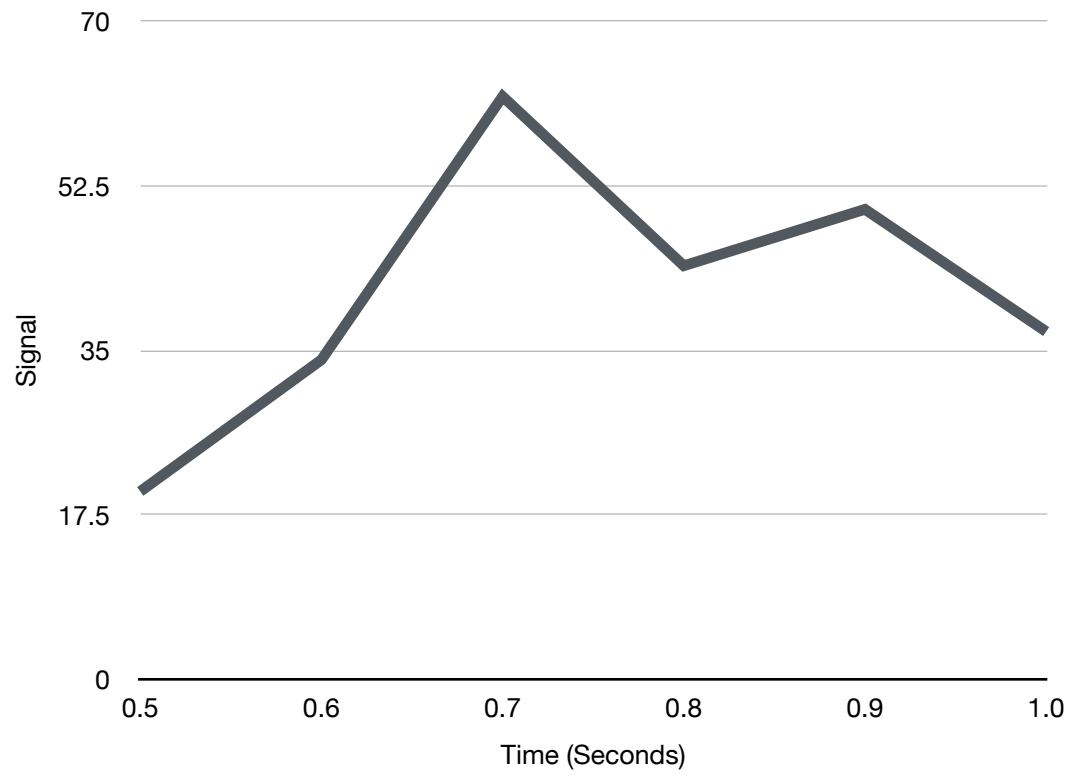


Figure 2

Sample 3: Moderately disabled 46-year-old male with mild injury

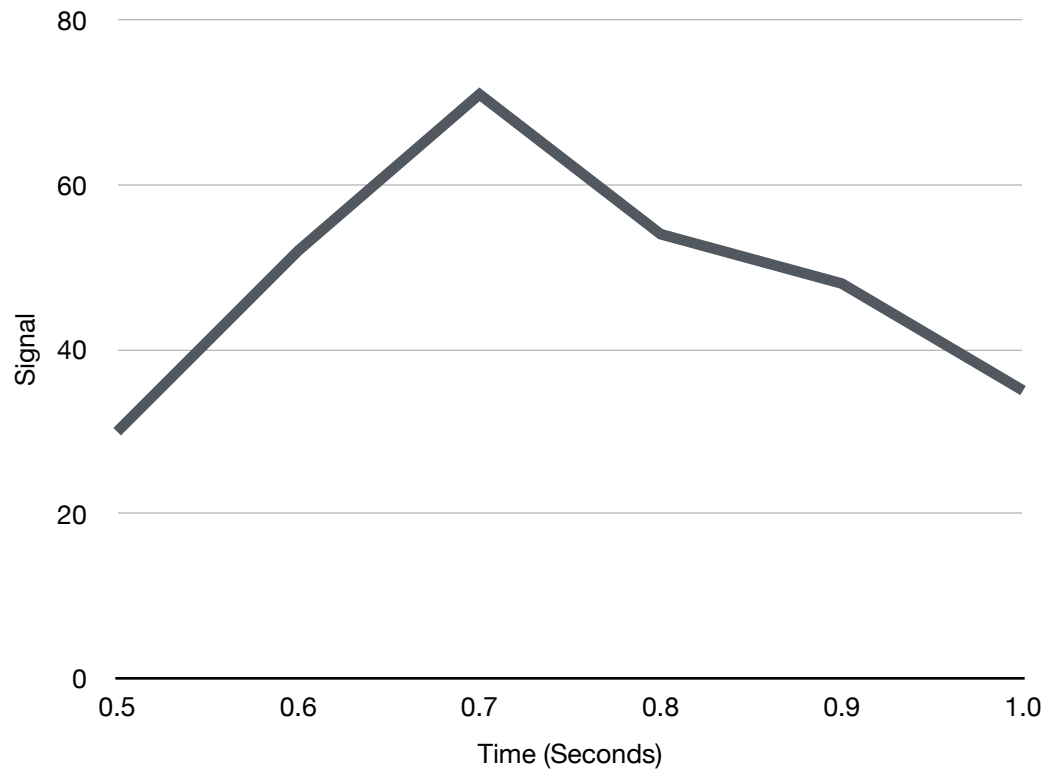


Figure 3

Taken with the data in Table 5, this suggests

Results

Forthcoming

